STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 11

RANDOM VARIABLES

DISCRETE RANDOM VARIABLE

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RANDOM VARIABLES

A random variable is formed by assigning a **numerical value** to each outcome in the sample space of a particular experiment. The state space of the random variable consists of these numerical values. Technically, a random variable can be thought of as being generated from a function that maps each outcome in a particular sample space onto the real number line R.

A random variable is a variable whose value is determined by the outcome of a random experiment.

*A **random variable** is a function that associates a real number with each element in the sample space.*

A **random variable** is a real-valued function whose domain is a sample space. ▪

Random variables are usually denoted by Capital letters *X, Y, Z* etc and their values are denoted by small letters *x, y, z* etc.

For example, the sample space giving a detailed description of each possible outcome when three electronic components are tested may be written

S = {NNN, NND, NDN, DNN, NDD, DND, DDN, DDD}

where *N* denotes non-defective and *D* denotes defective. One is naturally concerned with the number of defectives that occur. Thus, each point in the sample space will be *assigned a numerical value* of 0, 1, 2, or 3. These values are, of course, random quantities *determined by the outcome of the experiment*. They may be viewed as values assumed by the *random variable X*, the number of defective items when three electronic components are tested.

There are two types of random variables

1. Discrete Random Variable
2. Continuous Random Variable

DISCRETE RANDOM VARIABLE A random variable that assumes countable values is called discrete random variable.
For Example
1. Number of Students in a class
2. Number of Fish caught on a fishing trip

CONTINUIOUS RANDOM VARIABLE A random variable that can assume any value contained in one or more intervals is called continuous random variable. For Example
1. Weight of a Person
2. Temperature at a Place

PROBABILITY DISTRIBUTIONS

A Probability distribution gives the probability of each possible value of the variable.

DISCRETE PROBABILITY DISTRIBUTIONS

The Probability distribution of a discrete random variable lists all the possible values that the random variable can assume and their corresponding Probabilities. Simply A table or formula which consists all the values of a DRV with their respective Probabilities is known as Discrete Probability distribution.

Let $x_1, x_2, --, x_n$ be the values of a discrete random variable 'X' and $P(X=x_1), P(X=x_2), ----, P(X=x_n)$ are the probabilities of the values than the following table is known as Probability distribution of RV 'X'.

X	P(X = x)
x_1	$P(X=x_1)$
x_2	$P(X=x_2)$
x_3	$P(X=x_3)$
⋮	⋮
x_n	$P(X=x_n)$

Note that

① $P(X=x) \geq 0$

② $\sum_x P(X=x) = 1$

③ $P(X=x) = f(x)$ in DRV.

④ Probability dist. is also known as Probability Mass Function or Discrete Prob. Distribution.

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EXAMPLES OF DISCRETE PROBABILITY DISTRIBUTIONS

EXAMPLE① Find the Probability distribution of the Number of dots appeared when a die is rolled once.

SOLUTION when a die is rolled once then

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let 'X' denote the Number of dots appeared when a die is rolled once.

X	P(X=x)
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

which is the Required Prob-distribution of X.

clearly $\sum P(X=x) = 1$

EXAMPLE② Find the Probability distribution of the Number of Heads obtained when two fair coins are tossed once.

SOLUTION when TWO coins are tossed once

$$S = \{HH, HT, TH, TT\}$$

Let 'X' denote the Number of Heads appeared.

X	P(X=x)
0	1/4
1	2/4
2	1/4

which is the Probability distribution of X.

clearly $\sum P(X=x) = 1$

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EXAMPLE ③ Find the Probability distribution of Sum of dots appeared when two dice are thrown once.

SOLUTION when two dice are thrown once. than

$S = \{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),(2,1),(2,2),(2,3),(2,4),(2,5),(2,6),$
$(3,1),(3,2),(3,3),(3,4),(3,5),(3,6),(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)$
$(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}$

Let x denote the Sum of dots appeared when two dice are thrown once.

x	P(x=x)
2	1/36
3	2/36
4	3/36
5	4/36
6	5/36
7	6/36
8	5/36
9	4/36
10	3/36
11	2/36
12	1/36

which is the required Probability distribution of X.

clearly

$\sum P(x=x) = 1$

EXAMPLE ④ A committee of 3 persons is selected from 3 boys and 4 Girls. Construct a Probability distribution for the Number of Boys on the committee.

SOLUTION Here Boys = 3, Girls = 4, Select = 3.

Total = 7

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Let 'X' denote the Number of Boy's selected for committee.

X	P(X=x)
0	$\frac{^3C_0 \times {}^4C_3}{^7C_3} = 4/35$
1	$\frac{^3C_1 \times {}^4C_2}{^7C_3} = 18/35$
2	$\frac{^3C_2 \times {}^4C_1}{^7C_3} = 12/35$
3	$\frac{^3C_3 \times {}^4C_0}{^7C_3} = 1/35$

which is the Required Probability Distribution of X. clearly $\sum P(X=x) = 1$

EXAMPLE ⑤ The Probability Mass function of random variable 'X' is given by $P(X=x) = cx^2$, for $x = 0, 1, 2, 3, 4$. The find.

(a) the value of c.

(b) $P(X>2)$ (c) $P(X \leq 3)$ (d) $P(1<X<4)$ (e) $P(0 \leq X \leq 2)$.

SOLUTION.

X	P(X=x)	P(X=x)
0	0	0
1	c	1/30
2	4c	4/30
3	9c	9/30
4	16c	16/30

(a) value of c = ?

As $\sum P(X=x) = 1$

$0 + c + 4c + 9c + 16c = 1$

$30c = 1$

$\boxed{c = 1/30}$

(b) $P(X>2)$

$= P(X=3) + P(X=4)$

$= \frac{9}{30} + \frac{16}{30} = \frac{25}{30} = \frac{5}{6}$

(c) $P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3) = 0 + \frac{1}{30} + \frac{4}{30} + \frac{9}{30} = \frac{14}{30}$

(d) $P(1<X<4) = P(X=2) + P(X=3) = \frac{4}{30} + \frac{9}{30} = \frac{13}{30}$

(e) $P(0 \leq X \leq 2) = P(X=0) + P(X=1) + P(X=2) = 0 + \frac{1}{30} + \frac{4}{30} = \frac{5}{30} = \frac{1}{6}$

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DISTRIBUTION FUNCTIONS

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THE CUMULATIVE DISTRIBUTION FUNCTION

In many problems, we may be interested to know the probability that the value of random variable 'X' is less than or equal to some real number x. In this case we are looking at cumulative Probabilities ie for any random variable X, we may look at $P(X \leq b)$ for any real number b.

DEFINITION The distribution Function for a Random Variable 'X' evaluated at 'b' is denoted by $F(b)$ and is given by $F(b) = P(X \leq b)$

Note that distribution Function is also called the cumulative distribution Function.

PROPERTIES OF DISTRIBUTION FUNCTION

Every distribution function must satisfy four Properties given below

① $\lim_{x \to -\infty} F(x) = 0$

② $\lim_{x \to \infty} F(x) = 1$

③ The distribution function is a Non-decreasing function. ie if $a < b$, $F(a) \leq F(b)$.

④ The distribution function is Right-Hand continuous ie $\lim_{h \to 0^+} F(x+h) = F(x)$

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DISTRIBUTION FUNCTION FOR DISCRETE RANDOM VARIABLE

If 'x' is a discrete Random Variable. then

$$F(b) = P(X \leq b)$$

$$F(b) = \sum_{x=-\infty}^{b} p(x)$$

where $p(x)$ is the probability function.

Finding Probabilities using CDF.

① $P(X > a) = 1 - P(X \leq a) = 1 - F(a)$

② $P(X < a) = P(X \leq a-1) = F(a-1)$

③ $P(X \leq a) = F(a)$

④ $P(X \geq a) = 1 - P(X < a) = 1 - P(X \leq a-1) = 1 - F(a-1)$

⑤ $P(a \leq X \leq b) = F(b) - F(a-1)$

⑥ $P(a < X < b) = F(b-1) - F(a)$

⑦ $P(X = a) = F(a) - F(a-1)$

EXAMPLES

EXAMPLE① Find distribution function when two coins are tossed once.

SOLUTION Here $S = \{HH, HT, TH, TT\}$

let 'x' denote Number of Heads.

x	f(x)=P(x)	F(x)
0	1/4	1/4
1	2/4	3/4
2	1/4	1
	1	

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EXAMPLE-② Find the Distribution Function of the number of dots appeared when a fair die is rolled once.

SOLUTION Here $S = \{1, 2, 3, 4, 5, 6\}$

Let 'x' denote the dots appeared.

X	P(x)	F(x)
1	1/6	1/6
2	1/6	2/6
3	1/6	3/6
4	1/6	4/6
5	1/6	5/6
6	1/6	6/6

EXAMPLE-③ Find the Distribution Function of sum of dots appeared, when two dice are thrown once.

SOLUTION Here $S = \{(1,1), (1,2), \ldots, (6,6)\}$

Let 'x' denote sum of dots appeared.

X	P(x)	F(x)
2	1/36	1/36
3	2/36	3/36
4	3/36	6/36
5	4/36	10/36
6	5/36	15/36
7	6/36	21/36
8	5/36	26/36
9	4/36	30/36
10	3/36	33/36
11	2/36	35/36
12	1/36	36/36

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EXAMPLE-④ For a discrete Random variable 'x' the cumulative distribution function F(x) is given as

x	1	2	3	4	5
F(x)	0.2	0.32	0.67	0.9	1

Find the Probabilities using CDF

(a) $P(X=3)$

(b) $P(X>2)$

(c) $P(X<4)$

SOLUTION (a) $P(X=3) = F(3) - F(2) = 0.67 - 0.32 = 0.35$

(b) $P(X>2) = 1 - P(X \leq 2) = 1 - F(2) = 1 - 0.32 = 0.68$

(c) $P(X<4) = P(X \leq 3) = F(3) = 0.67.$

EXAMPLE⑤ The cumulative Probabilities for a Random variable 'x' are given in the following table, use table to find

x	F(x)
0	0.0388
1	0.1756
2	0.4049
3	0.6477
4	0.8298
5	0.9327
6	0.9781
7	0.9941
8	0.9987
9	0.9998
10	1

(a) $P(X \leq 5)$

(b) $P(X>3)$

(c) $P(3 \leq X \leq 7)$

(d) $P(X=7)$

(e) $P(X \geq 8)$

SOLUTION

(a) $P(X \leq 5) = F(5) = 0.9327$

(b) $P(X \geq 3) = 1 - F(3) = 1 - 0.6477 = 0.3523$

(c) $P(3 \leq X \leq 7) = F(7) - F(2) = 0.9941 - 0.4049 = 0.5892$

(d) $P(X=7) = F(7) - F(6) = 0.9941 - 0.9781 = 0.016$

(e) $P(X \geq 8) = 1 - P(X<8) = 1 - F(7) = 1 - 0.9941 = 0.0059$

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MATHEMATICAL EXPECTATION.

It is an important concept in Probability and Statistics. If X is a Random variable than expectation of X is denoted by E(X) and it gives an average or Mean value of X.
In general it gives a quick picture of the long-run 'Average' result when the experiment is performed over and over again.

PROPERTIES OF EXPECTATION

If X is a random variable and a, b are constants than the following properties hold.

① $E(a) = a$
② $E(aX) = a \cdot E(X)$
③ $E(aX+b) = aE(X)+b$

EXPECTATION OF DISCRETE RANDOM VARIABLES

If 'X' is a discrete random variable, than the Mean or Expected value of x is given by

$$E(X) = \sum x f(x)$$

If 'X' is a discrete random variable with probability function $f(x)$, than the expectation of the function $g(x)$ is given by

$$E(g(x)) = \sum_x g(x) f(x)$$

Note that

① $E(x) = \sum x f(x)$
② $E(x^2) = \sum x^2 f(x)$
③ $E(x^3) = \sum x^3 f(x)$

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EXAMPLES

EXAMPLE-① Find the expected Number of Heads when two fair coins are tossed.

SOLUTION Let 'x' denote the Number of Heads when two fair coins are tossed.

x	p(x)	x p(x)
0	1/4	0
1	2/4	2/4
2	1/4	1/2

So $E(x) = \sum x\,p(x)$

$= 0 + \frac{2}{4} + \frac{1}{2}$

$E(x) = 1$

EXAMPLE-② Find the expected Number of Sixes when three fair dice are thrown.

SOLUTION Let 'x' denote the Number of Sixes

So 'x' can take the values 0, 1, 2, 3

Now let D_1 denote Six on the 1st die

D_2 " " " 2ND die.

D_3 " " " 3rd die.

Also $P(D_1) = \frac{1}{6}$, $P(D_1') = \frac{5}{6}$

So $P(X=0) = P(D_1' \cap D_2' \cap D_3') = (\frac{5}{6})(\frac{5}{6})(\frac{5}{6}) = \frac{125}{216}$

$P(X=1) = P(D_1 \cap D_2' \cap D_3') + P(D_1' \cap D_2 \cap D_3') + P(D_1' \cap D_2' \cap D_3)$

$= (\frac{1}{6})(\frac{5}{6})(\frac{5}{6}) + (\frac{5}{6})(\frac{1}{6})(\frac{5}{6}) + (\frac{5}{6})(\frac{5}{6})(\frac{1}{6})$

$= \frac{25}{216} \times 3 = \frac{75}{216}$

$P(X=2) = P(D_1 \cap D_2 \cap D_3') + P(D_1 \cap D_2' \cap D_3) + P(D_1' \cap D_2 \cap D_3)$

$= (\frac{1}{6})(\frac{1}{6})(\frac{5}{6}) + (\frac{1}{6})(\frac{5}{6})(\frac{1}{6}) + (\frac{5}{6})(\frac{1}{6})(\frac{1}{6})$

$= \frac{5}{216} \times 3 = \frac{15}{216}$

$P(X=3) = P(D_1 \cap D_2 \cap D_3) = \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \frac{1}{216}$

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Now Expectation is given by the table.

x	0	1	2	3
P(x)	125/216	75/216	15/216	1/216
xP(x)	0	75/216	30/216	3/216

Now $E(x) = \sum x P(x)$

$$E(x) = 0 + \frac{75}{216} + \frac{30}{216} + \frac{3}{216} = \frac{108}{216} = 0.5$$

EXAMPLE-③ The Probability distribution of a random variable 'x' is shown in the table.

x	1	2	3	4	5
P(x=x)	0.1	0.3	c	0.2	0.1

Find the following

(a) value of 'c'.

(b) E(x)

(c) $E(x^2)$

(d) E(3)

(e) E(2x)

(f) E(2x+3)

SOLUTION

(a) WKT $\sum P(x) = 1$

$$0.1 + 0.3 + c + 0.2 + 0.1 = 1$$

$$c = 1 - 0.7$$

$$\boxed{c = 0.3}$$

(b) Find E(x).

x	1	2	3	4	5
P(x)	0.1	0.3	0.3	0.2	0.1
xP(x)	0.1	0.6	0.9	0.8	0.5

So E(x)
$= 0.1 + 0.6 + 0.9 + 0.8 + 0.5$
$= 2.9$

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(c) For $E(x^2)$

$E(x^2) = \sum x^2 P(x)$

$= 9.7$

x	$P(x)$	x^2	$x^2P(x)$
1	0.1	1	0.1
2	0.3	4	1.2
3	0.3	9	2.7
4	0.2	16	3.2
5	0.1	25	2.5
			9.7

(d) $E(3)$

$= \sum 3 \cdot P(x)$

$= 3 \sum P(x)$

$= 3(1) = 3$

(e) $E(2x)$

$= \sum 2x P(x)$

$= 2 E(x)$

$E(2x) = 2 E(x)$

$= 2(2.9)$

$= 5.8$

(f) $E(2x+3)$

$= 2E(x) + 3$

$= 2(2.9) + 3$

$= 5.8 + 3$

$= 8.8$

or $E(2x+3) = \sum (2x+3) \cdot P(x) = 8.8$.

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VARIANCE AND STANDARD DEVIATION

If 'x' is a random variable then its variance is given by the formula

$$Var(x) = E(x-\mu)^2$$

or $$Var(x) = E(x^2) - \mu^2$$

or $$Var(x) = E(x^2) - (E(x))^2$$

PROOF $Var(x) = E(x-\mu)^2$

$= E(x^2 + \mu^2 - 2\mu E(x))$

$= E(x^2) + E(\mu^2) - 2\mu E(x)$

$= E(x^2) + \mu^2 - 2\mu.\mu$

$= E(x^2) - \mu^2$

$= E(x^2) - (E(x))^2$

PROPERTIES OF VARIANCE.

If 'x' is a variable and a, b are constants. then

1) $Var(a) = 0$
2) $Var(ax) = a^2 . Var(x)$
3) $Var(ax+b) = a^2 . Var(x)$.

STANDARD DEVIATION

If 'x' is a random variable then Standard deviation of x is denoted by 'σ' and is given by

$$\sigma = \sqrt{Var(x)}$$

EXAMPLES

EXAMPLE-④ The random variable 'x' has probability distribution as shown in the table.

Find (a) E(x)
(b) $E(x^2)$
(c) Var(x)
(d) S.D(x)

x	1	2	3	4	5
P(x=x)	0.1	0.3	0.2	0.3	0.1

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SOLUTION consider the table.

x	$P(x)$	$xP(x)$	x^2	$x^2P(x)$
1	0.1	0.1	1	0.1
2	0.3	0.6	4	1.2
3	0.2	0.6	9	1.8
4	0.3	1.2	16	4.8
5	0.1	0.5	25	2.5
	1	3		10.4

Now

(a) $E(x) = \Sigma xP(x) = 3$

(b) $E(x^2) = \Sigma x^2P(x) = 10.4$

(c) $Var(x) = E(x^2) - (E(x))^2$

$= 10.4 - (3)^2$

$= 10-4-9 = 1.4$

(d) $S.D(x) = \sigma = \sqrt{V(ar(x)} = \sqrt{1.4} = 1.18$

EXAMPLE-⑤ 'x' is the random variable 'the Number on a biased die and the p.d.f of x is as shown

x	1	2	3	4	5	6
$P(x)$	1/6	1/6	1/5	c	1/5	1/6

Find the following

(a) value of 'c'

(b) $E(x)$

(c) $E(x^2)$

(d) $Var(x)$

(e) $S.D(x)$

(f) $E(2x+3)$

(g) $V(3x+4)$

(h) $S.D(4x+5)$

SOLUTION (a) WKT $\Sigma P(x) = 1$

$\frac{1}{6} + \frac{1}{6} + \frac{1}{5} + c + \frac{1}{5} + \frac{1}{6} = 1$ so $\boxed{c = \frac{1}{10}}$

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SOLUTION. consider the table.

x	$P(x)$	$x \cdot P(x)$	x^2	$x^2 \cdot P(x)$
1	1/6	1/6	1	1/6
2	1/6	2/6	4	4/6
3	1/5	3/5	9	9/5
4	1/10	4/10	16	16/10
5	1/5	5/5	25	25/5
6	1/6	6/6	36	36/6
	1	3.5		457/30

(b) $E(x) = \Sigma x P(x)$

$= 3.5$

(c) $E(x^2) = \Sigma x^2 P(x)$

$= \frac{457}{30}$

(d) $Var(x) = E(x^2) - (E(x))^2$

$= \frac{457}{30} - (3.5)^2 = \frac{457}{30} - \frac{49}{4} = \frac{179}{60} = 2.983$

(e) $S.D(x) = \sqrt{Var(x)} = 1.7272$

(f) $E(2x+3)$

$= 2E(x)+3$

$= 2(3.5)+3 = 10$

(g) $V(3x+4)$

$= 9 \cdot V(x) = 9 \times \frac{179}{60} = \frac{1611}{60}$

(h) $S.D(4x-5)$

For S.D we shall first find $Var(4x-5)$

$Var(4x-5) = 16 \cdot V(x) = 16 \cdot \frac{179}{60} = \frac{716}{15}$

Now $S.D(4x-5) = \sqrt{Var(4x-5)} = \sqrt{\frac{716}{15}} = 6.91$

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